1 Question (10 points)

$$
\begin{aligned}
E &\rightarrow E + E \\
E &\rightarrow E * E \\
E &\rightarrow (E) \\
E &\rightarrow a
\end{aligned}
$$

For the given CFG above. Is the grammar ambiguous? Show it.

2 Question (15 points)

Convert the following grammar into CNF.

$$
\begin{aligned}
S &\rightarrow SA \\
S &\rightarrow 1 \\
S &\rightarrow AAA \\
A &\rightarrow 0A1 \\
A &\rightarrow S \\
A &\rightarrow 0 \\
A &\rightarrow \epsilon
\end{aligned}
$$

3 Question (15 points)

Is the following language context-free? Prove your answer.

$B = \{a^{n-m}b^n c^m | n \geq m \geq 0\}$

4 Question (10 points)

Given the context free language C = $\{a^p b^m c^n d^n e^m f^p | n, m, p \geq 0\}$. Try do disprove it by pumping lemma with the string, $a^p b^p c^p d^p e^p f^p$, and give two general example cases where you can successfully pump up, and two general example cases where you can not successfully pump up. In each case, select strings v and y consisting of only one distinct character.